

EVCharge Android Application

APK Build & Deployment Guide

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For internal use — to be read before building the installable APK

System Overview

The EVCharge application is an Android mobile app that communicates with a Node.js/Express backend server running on a PC. The app and the server must be on the **same network** for the app to function. The server runs on **port 3001**.

Component	Details
Android App	EVCharge — built with Kotlin + Jetpack Compose
Backend Server	Node.js + Express.js, running on the PC at port 3001
Database	MySQL 8, running locally on the PC
Communication	HTTP REST API + WebSocket (ws://)
Push Notifications	Firebase Cloud Messaging (FCM) — requires internet

The Critical Configuration File

There is **one file** that controls how the app connects to the server:

```
EVCharge_app/app/src/main/java/com/example/evcharge/network/ServerConfig.kt
```

Its content is:

```
object ServerConfig {  
    const val HOST      = "192.168.137.1"    // <-- THIS IS WHAT YOU CHANGE  
    const val PORT      = 3001
```

```
const val HTTP_BASE = "http://$HOST:$PORT/"  
const val WS_URL    = "ws://$HOST:$PORT/ws"  
}
```

Important: The `HOST` value must match the **PC's IP address** on the network the phone will use. If the IP is wrong, the map will be empty and no data will load. This is the only value that ever needs to change.

Current state of the file: At the time of writing, `ServerConfig.kt` contains `HOST = "192.168.100.3"` (used for home testing on the local WiFi network). **Before building the presentation APK, this value must be changed to "192.168.137.1"** so the app connects via the Windows hotspot.

Scenario A — Presentation / Demo Day (Recommended Setup)

This is the **recommended setup for the presentation**. The PC creates its own WiFi hotspot, so no external router is needed and the IP is always the same.

When Windows creates a mobile hotspot, it always assigns itself the IP address **192.168.137.1** — this never changes regardless of location.

Step 1 — Set the `HOST` in `ServerConfig.kt`

Make sure `ServerConfig.kt` contains:

```
const val HOST = "192.168.137.1"
```

Step 2 — Enable Windows Mobile Hotspot

1. Open **Settings** → **Network & Internet** → **Mobile hotspot**
2. Turn the hotspot **ON**
3. Note the network name (SSID) and password shown on screen

Step 3 — Connect the Android Phone

1. On the Android phone, open **WiFi settings**
2. Connect to the hotspot network created in Step 2
3. Confirm the phone shows "Connected"

Step 4 — Start the Backend Server

On the PC, open a terminal in the server folder and run:

```
cd server  
npm start
```

Wait until the terminal shows the server is running on port 3001.

Step 5 — Launch the App

Open the EVCharge app on the phone. The map should load and stations should appear.

Scenario B — Testing at Home (Same WiFi Router)

If the PC and phone are both connected to the same home WiFi router:

Step 1 — Find the PC's IP Address

Open a terminal on the PC and run:

```
ipconfig
```

Look for **Adresse IPv4** under the WiFi adapter. Example: 192.168.100.3

Step 2 — Update ServerConfig.kt

Replace the HOST value with the IP found above:

```
const val HOST = "192.168.100.3" (use your actual IP)
```

Step 3 — Start the Server and Test

Same as Scenario A Steps 4 and 5.

Home WiFi limitation: The PC's IP from the router can change each time the PC restarts or reconnects. If stations stop appearing, re-run `ipconfig`, check the IP, and update `ServerConfig.kt`. This is why Scenario A (hotspot) is preferred for the presentation.

Building the Installable APK

Before building: Make sure `ServerConfig.kt` has the correct HOST for the target scenario (Scenario A for presentation: `192.168.137.1`).

Step 1 — Open the Project in Android Studio

Open Android Studio and open the folder:

```
MFE_V2/EVCharge_app/
```

Wait for Gradle sync to complete (progress bar at the bottom).

Step 2 — Select Build Variant

In Android Studio, go to:

Build → **Select Build Variant** → choose **release**

Step 3 — Generate Signed APK

1. Go to **Build** → **Generate Signed Bundle / APK**
2. Select **APK** and click **Next**
3. Select or create a keystore file (keep it safe — needed for future updates)
4. Fill in key alias, key password, and store password
5. Click **Next**, select **release** build variant
6. Click **Finish**

The signed APK will be generated at:

```
EVCharge_app/app/release/app-release.apk
```

Step 4 — Install on the Android Device

Option A — Direct USB install from Android Studio:

1. Connect the phone via USB

2. Enable **Developer Options** and **USB Debugging** on the phone
3. Click the green **Run** button in Android Studio

Option B — Manual APK transfer:

1. Copy `app-release.apk` to the phone (via USB, email, or Google Drive)
2. On the phone, go to **Settings** → **Install unknown apps** and allow installation from the file manager
3. Open the APK file on the phone and tap **Install**

Pre-Demo Checklist

☐	ServerConfig.kt HOST is set to 192.168.137.1
☐	APK is built and installed on the Android device
☐	Windows Mobile Hotspot is enabled on the PC
☐	Phone is connected to the PC hotspot
☐	MySQL database is running on the PC
☐	Node.js server is started (<code>cd server && npm start</code>)
☐	App opens and stations appear on the map
☐	A test session can be started and stopped

Quick Reference — IP Values

Scenario	HOST value	When to use
Presentation (hotspot)	192.168.137.1	Demo day — always reliable
Home WiFi testing	192.168.100.3	Testing at home (may change)
Android Emulator	10.0.2.2	Running on PC emulator only

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